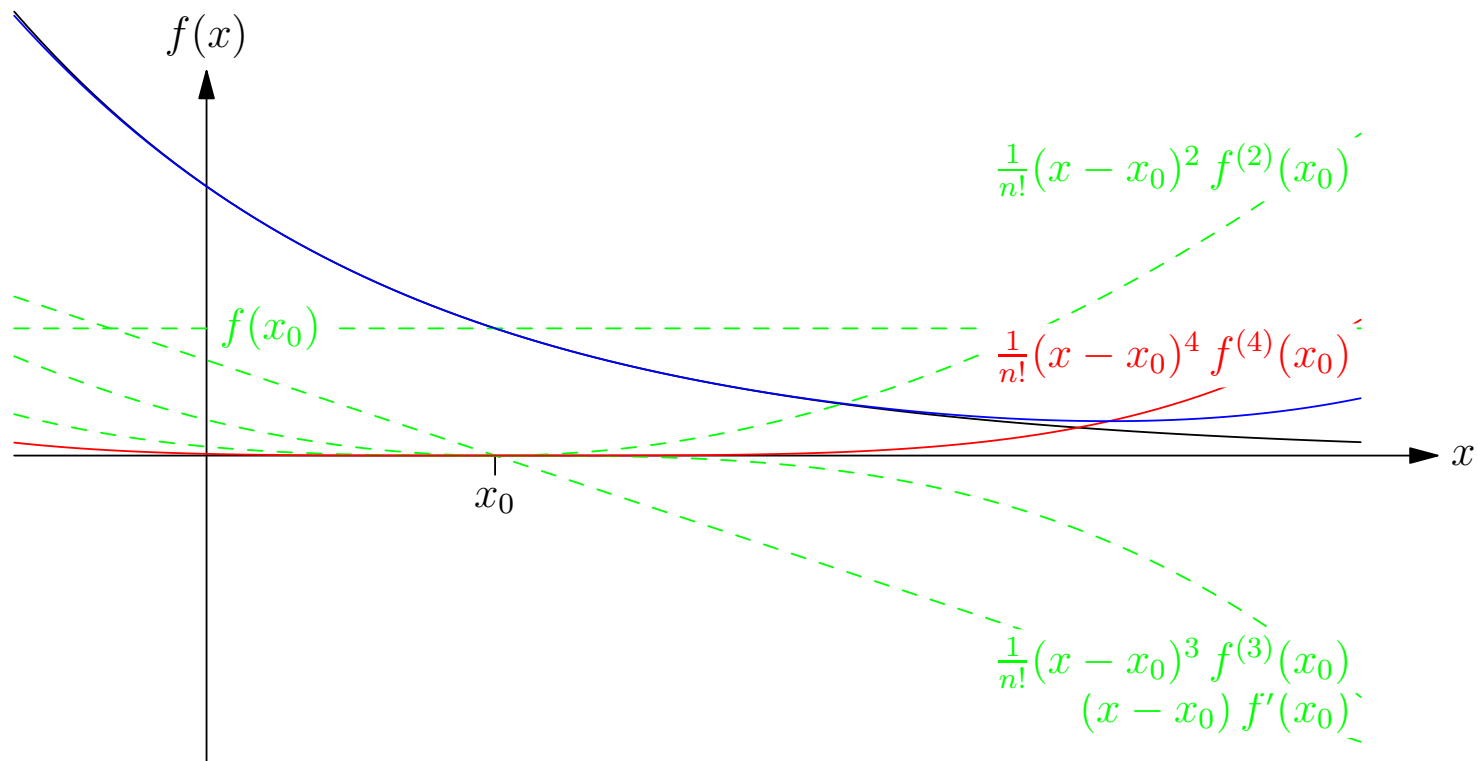


Mathematics of Machine Learning

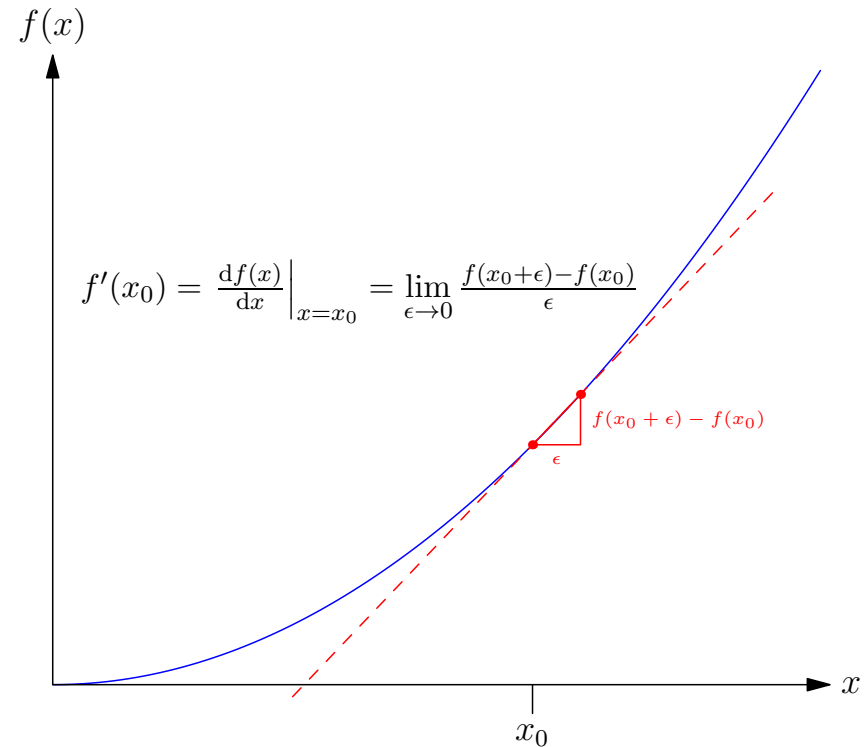
Differential Calculus



Differentiation, product and chain rules, vectors and matrices

Outline

1. **1-D Taylor Expansion?**
2. n -D Taylor Expansion?



Approximating Functions

- In much of machine learning we are trying to approximate functions
- There are many different ways to do this
- Depending on your application some are much better than others
- All function approximation makes some assumptions about the regularity of the function they are approximating
- Taylor expansions are polynomial approximations of a smooth function around a point

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Polynomial Approximation

- Suppose we wanted to approximate a function $f(x)$ around a point x_0 using a polynomial

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- Taking derivatives

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Taylor Expansion

- This gives the Taylor expansion

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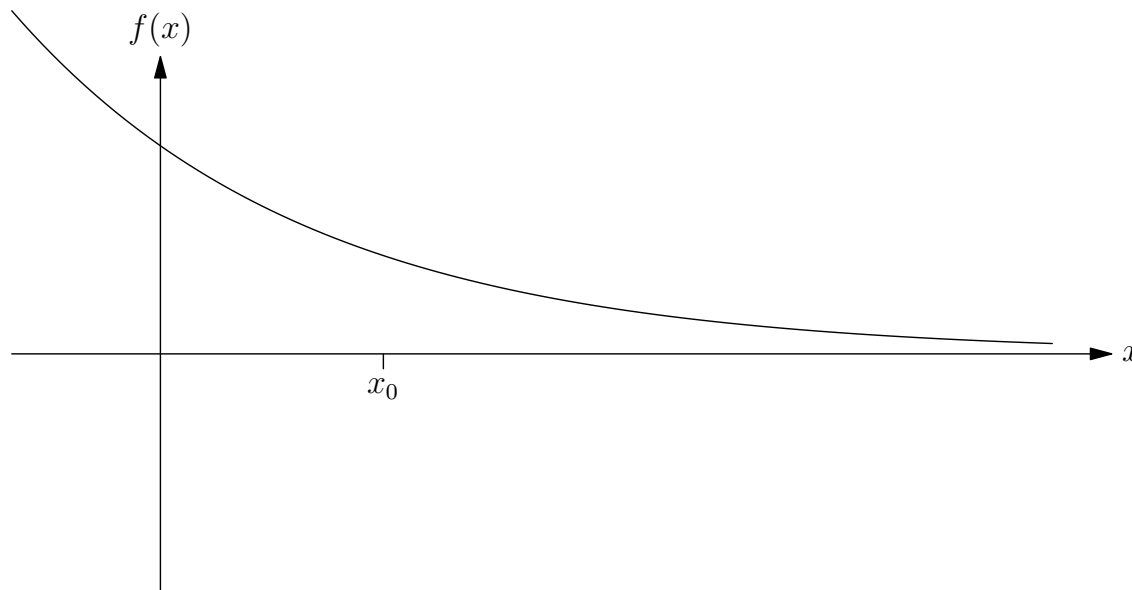
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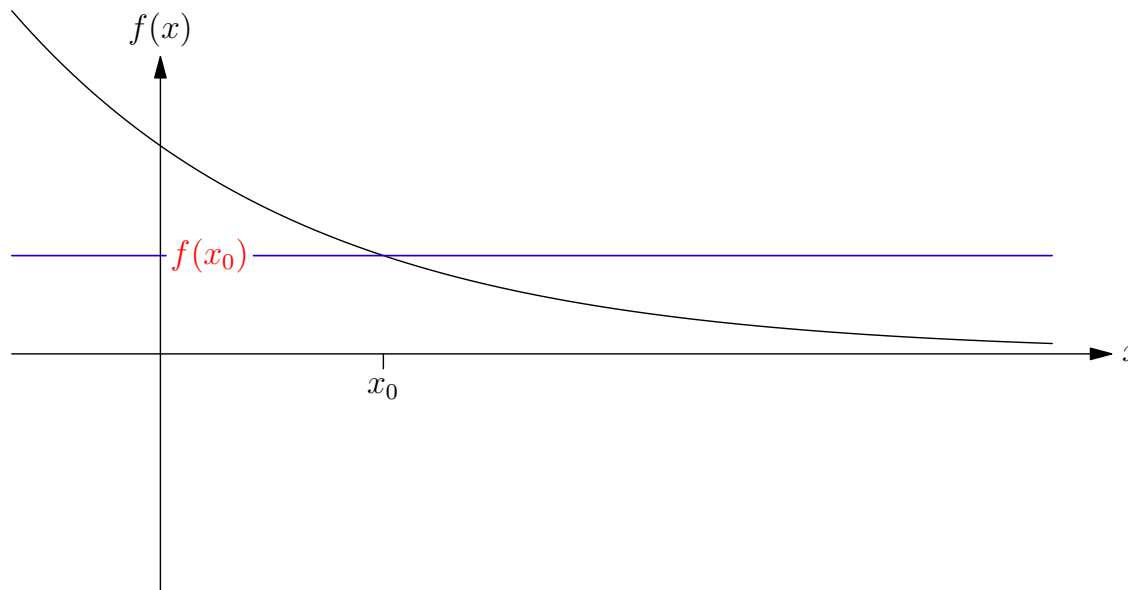


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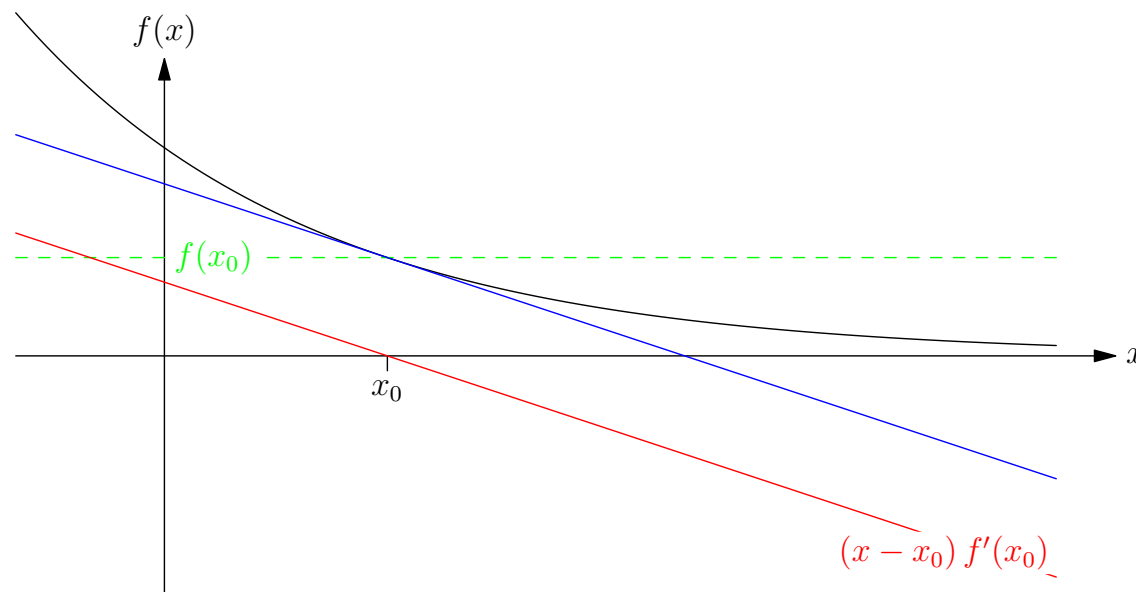


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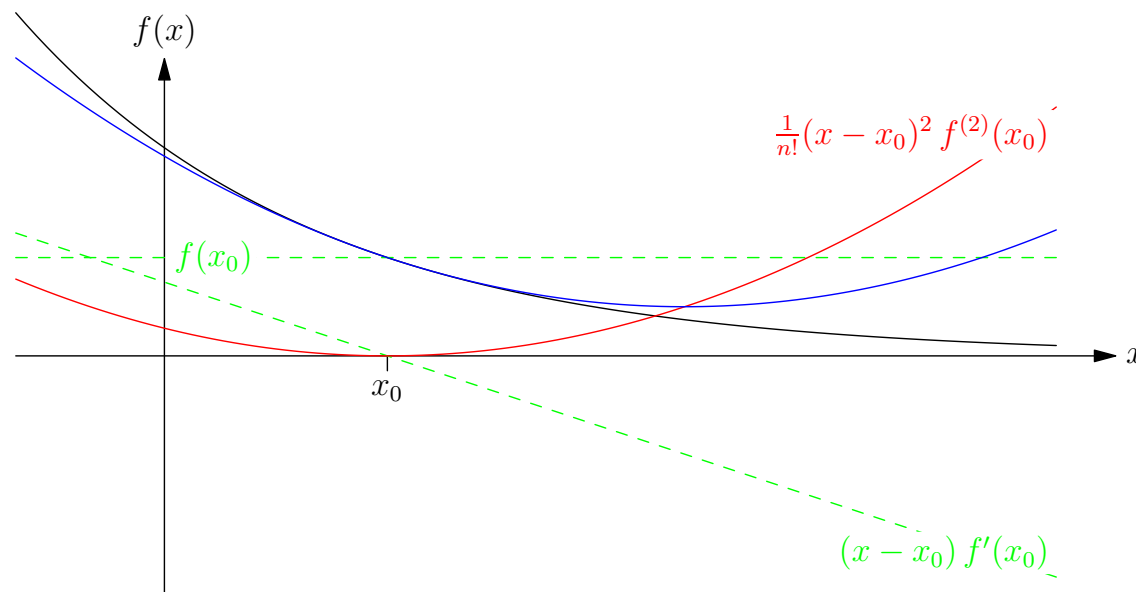


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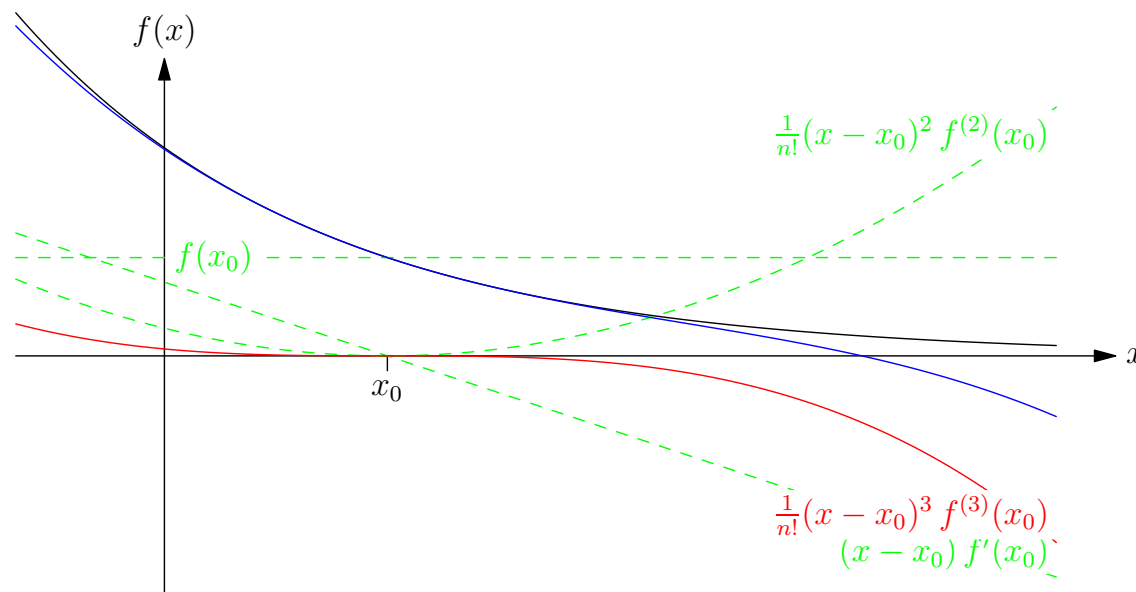


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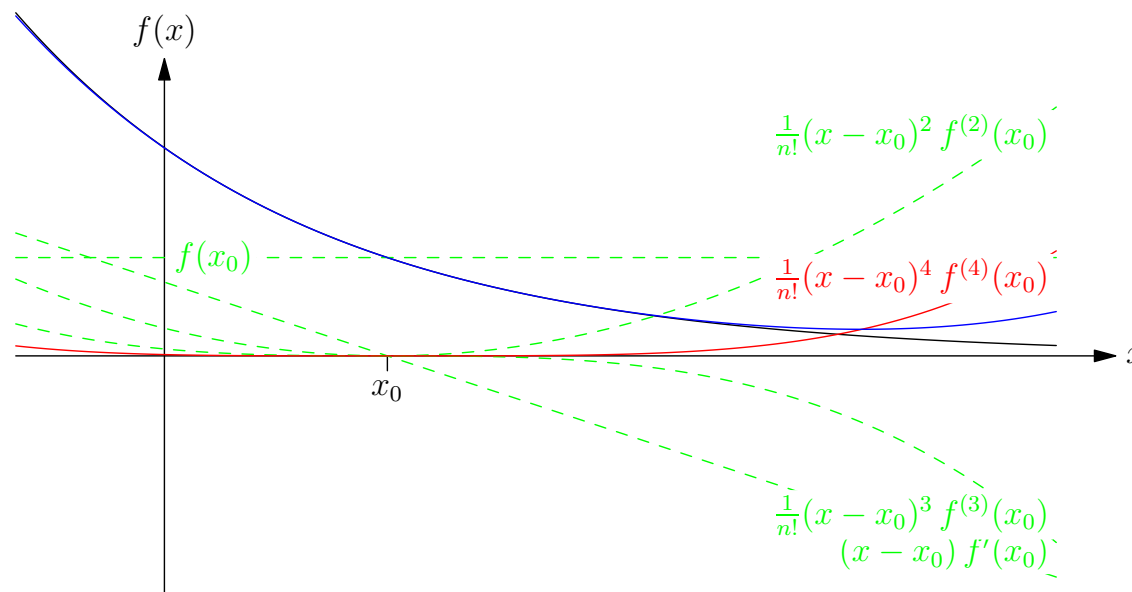


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Convergence

- As we add more and more terms our local approximation becomes more accurate, provided the higher derivatives exist
- Polynomials have the nice property that they only diverge at ∞
- However, polynomials cannot capture functions that diverge for finite numbers, e.g.

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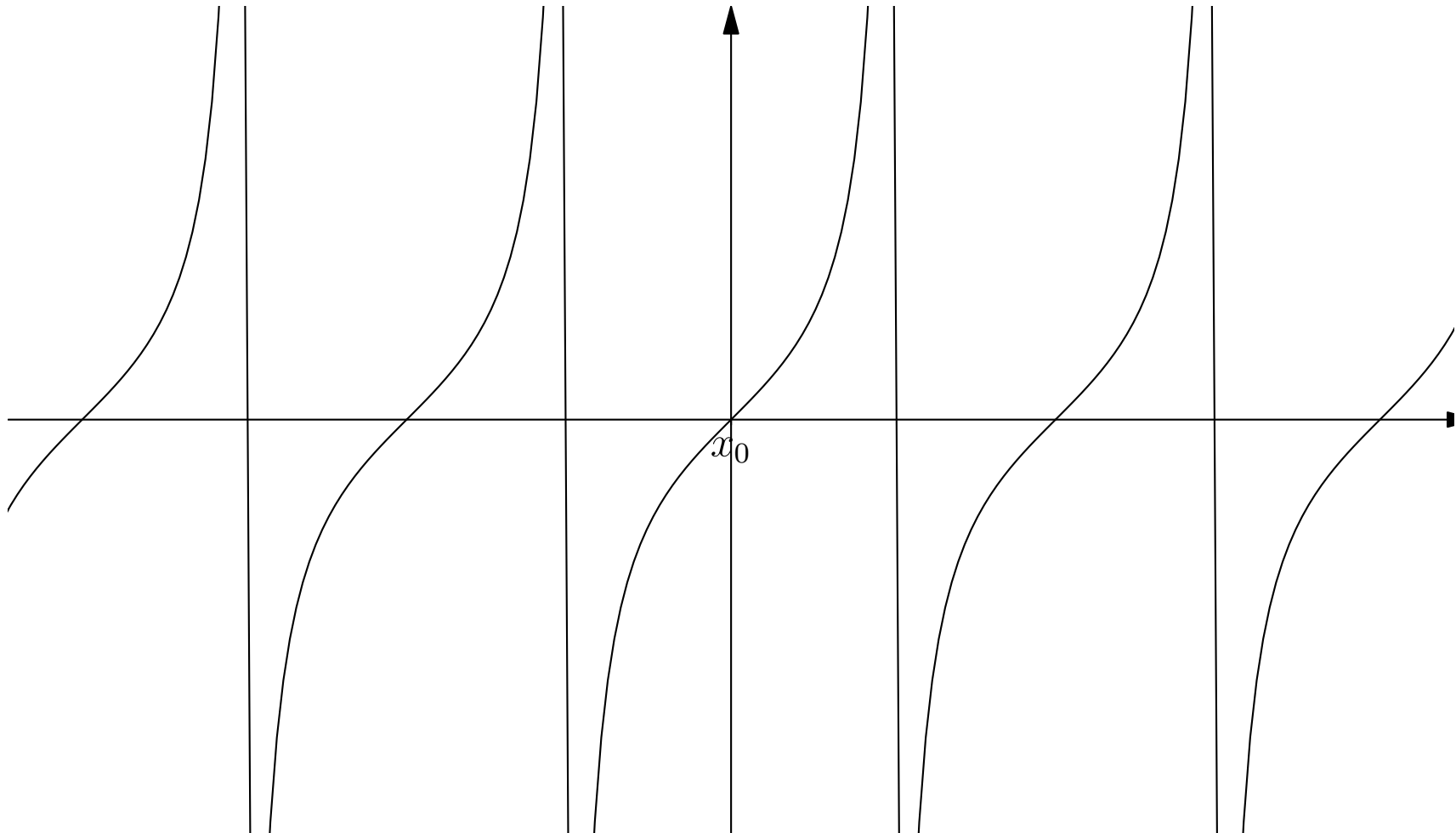
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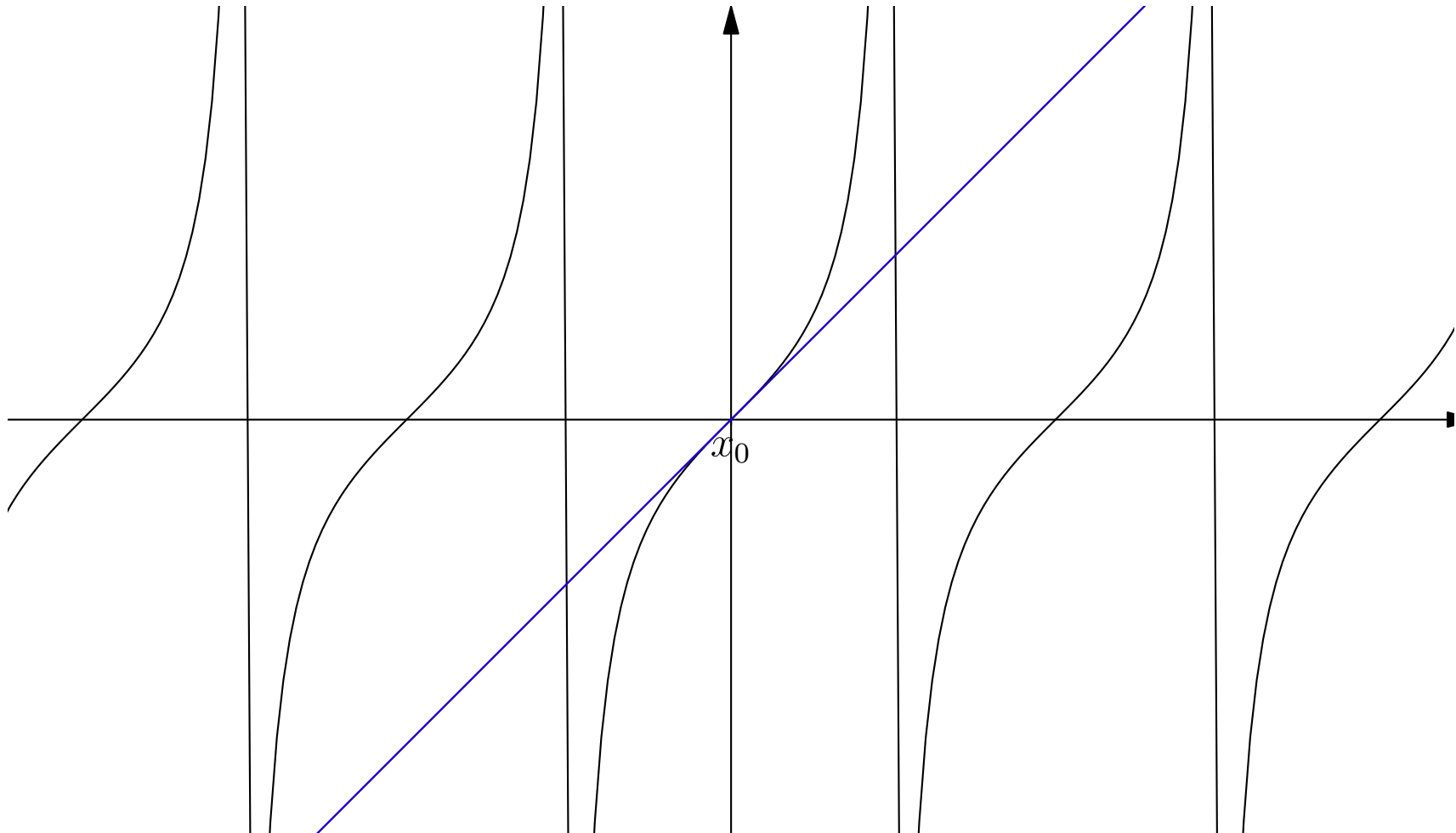
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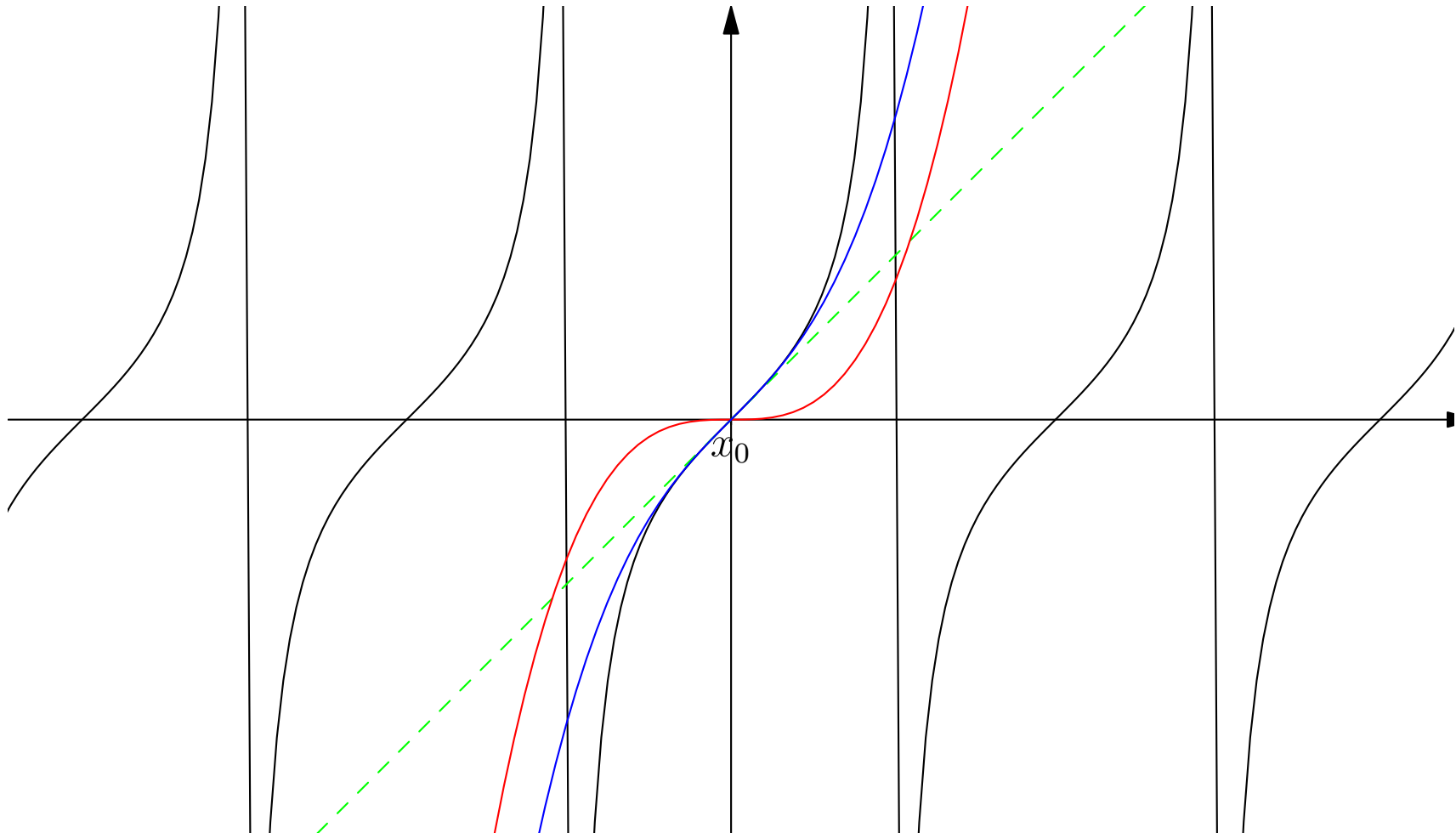
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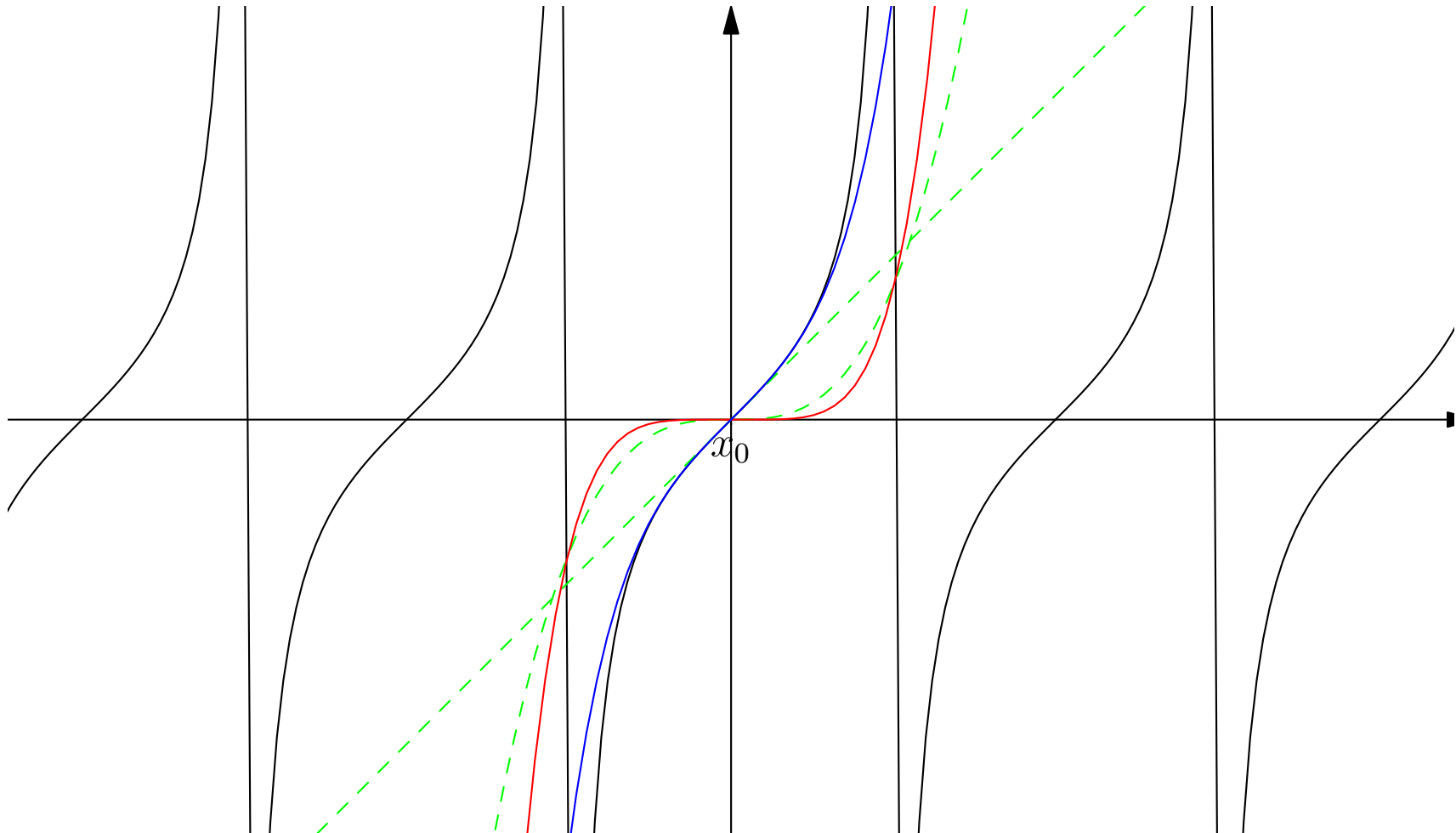
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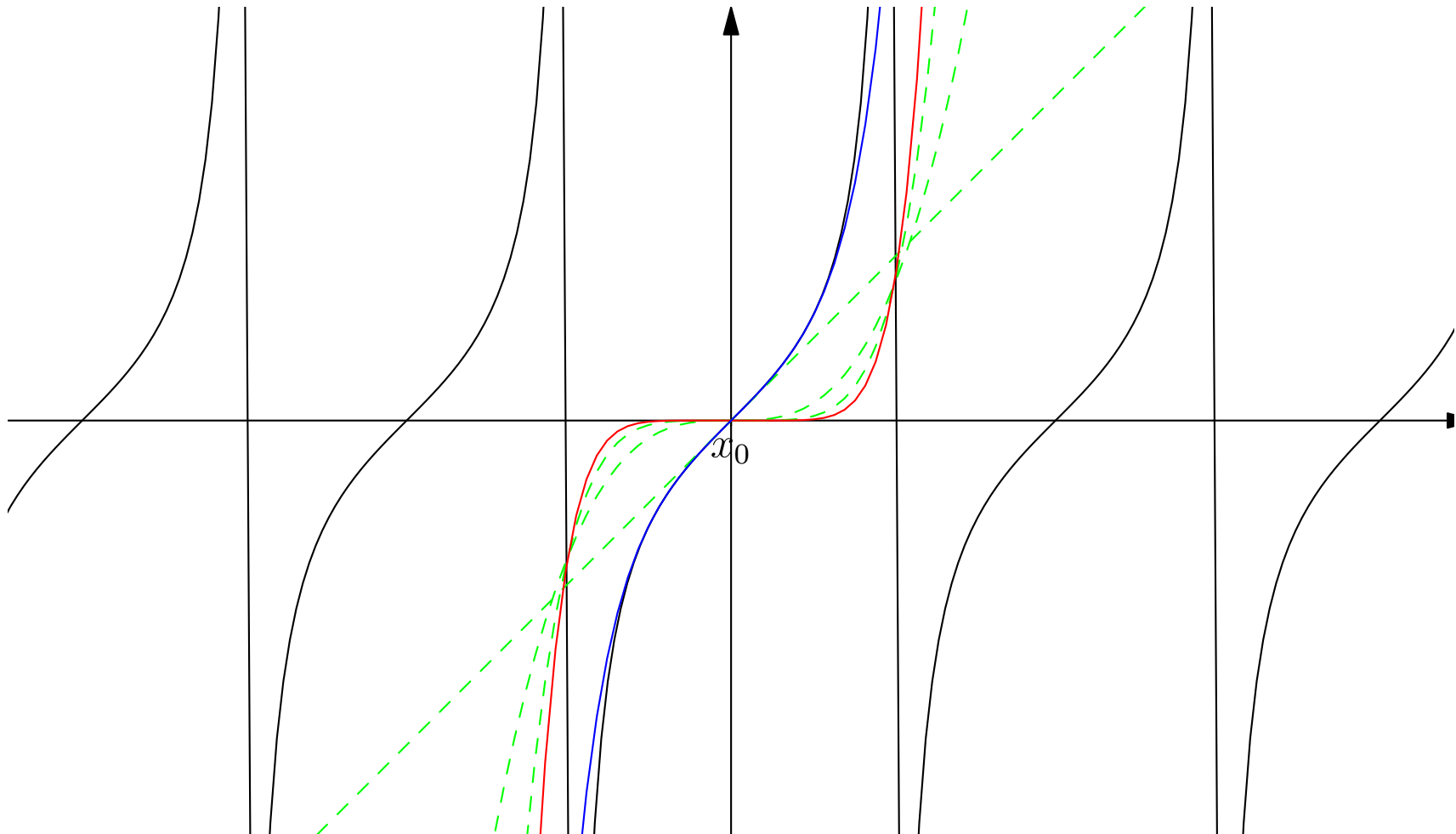
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